

1. Create user with name Techie and provide sudo access to user.

```
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ sudo useradd Techie
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ sudo passwd Techie
New password:
Retype new password:
passwd: password updated successfully
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ su Techie
Password:
$ id
uid=1002(Techie) gid=1002(Techie) groups=1002(Techie)
$ sudo apt install git
[sudo] password for Techie:
Techie is not in the sudoers file. This incident will be reported.
$ exit
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ find / -name sudoers
find: '/etc/polkit-1/localauthority': Permission denied
/etc/sudoers
```

```
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ sudo vi /etc/sudoers
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ su Techie
Password:
$ id
uid=1002(Techie) gid=1002(Techie) groups=1002(Techie)
$ sudo apt install git
[sudo] password for Techie:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libreoffice-ogltrans
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  git-man liberror-perl
Suggested packages:
  git-daemon-run | git-daemon-sysvinit git-doc git-email git-gui gitk gitweb git-cvs git-mediawiki git-svn
The following NEW packages will be installed:
  git git-man liberror-perl
0 upgraded, 3 newly installed, 0 to remove and 372 not upgraded.
Need to get 4,154 kB of archives.
After this operation, 21.0 MB of additional disk space will be used.
Do you want to continue? [Y/n] n
Abort.
$ exit
vamshikumar@vamshikumar-virtual-machine:~/Desktop$
```

```
#Defaults:%sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_proxy"

# This allows running arbitrary commands, but so does ALL, and it means
# different sudoers have their choice of editor respected.
#Defaults:%sudo env_keep += "EDITOR"

# Completely harmless preservation of a user preference.
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"

# While you shouldn't normally run git as root, you need to with etckeeper
#Defaults:%sudo env_keep += "GIT_AUTHOR_* GIT_COMMITTER_*"

# Per-user preferences; root won't have sensible values for them.
#Defaults:%sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"

# "sudo scp" or "sudo rsync" should be able to use your SSH agent.
#Defaults:%sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"

# Ditto for GPG agent
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL

# Members of the admin group may gain root privileges
%admin   ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL
Techie  ALL=(ALL:ALL) ALL
# See sudoers(5) for more information on "@include" directives:

@includedir /etc/sudoers.d
vamshikumar@vamshikumar-virtual-machine:~/Desktop$
```

2. Navigate to the home directory.

```
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ cd ~
vamshikumar@vamshikumar-virtual-machine:~$ ~
bash: /home/vamshikumar: Is a directory
vamshikumar@vamshikumar-virtual-machine:~$
```

3. Create a new directory.

```
vamshikumar@vamshikumar-virtual-machine: ~/Desktop
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ mkdir Techie_horizon
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ ls
f1.txt  new.txt  sabair  Techie_horizon
vamshikumar@vamshikumar-virtual-machine:~/Desktop$
```

4. List the contents of a directory.

```
vamshikumar@vamshikumar-virtual-machine: ~/Desktop
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ ls
f1.txt  new.txt  sabair  Techie_horizon
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ ll
total 28
drwxr-xr-x  5 vamshikumar vamshikumar 4096 Aug  7 16:29 ./
drwxr-x--- 17 vamshikumar vamshikumar 4096 Aug  7 13:06 ../
drw-rw-rw-  2 vamshikumar vamshikumar 4096 Aug  4 15:45 d2/
-rw-rw-r--  1 vamshikumar vamshikumar  11 Aug  6 17:00 f1.txt
-rw-rw-r--  1 vamshikumar vamshikumar   79 Aug  4 16:29 new.txt
dr--rwx--x  3 vamshikumar vamshikumar 4096 Aug  6 16:12 sabair/
drwxrwxr-x  2 vamshikumar vamshikumar 4096 Aug  7 16:29 Techie_horizon/
vamshikumar@vamshikumar-virtual-machine:~/Desktop$
```

5. Change the current directory.

```
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ cd ~
vamshikumar@vamshikumar-virtual-machine:~$ ~
bash: /home/vamshikumar: Is a directory
vamshikumar@vamshikumar-virtual-machine:~$
```

6. Create a new empty file.

```
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ touch test.txt
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ vi test.txt
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ cat test.txt
hello world
vamshikumar@vamshikumar-virtual-machine:~/Desktop$
```

7. View the contents of a file.

```
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ touch test.txt
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ vi test.txt
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ cat test.txt
hello world
vamshikumar@vamshikumar-virtual-machine:~/Desktop$
```

8. Copy a file to another location.

```
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ ll
total 32
drwxr-xr-x  5 vamshikumar vamshikumar 4096 Aug  7 16:48 ./
drwxr-xr-x 17 vamshikumar vamshikumar 4096 Aug  7 16:50 ../
drw-rw-rw-  2 vamshikumar vamshikumar 4096 Aug  4 15:45 .git/
-rw-rw-r--  1 vamshikumar vamshikumar  11 Aug  6 17:00 f1.txt
-rw-rw-r--  1 vamshikumar vamshikumar   79 Aug  4 16:29 new.txt
dr--rwx--x  3 vamshikumar vamshikumar 4096 Aug  6 16:12 sabair/
drwxrwxr-x  2 vamshikumar vamshikumar 4096 Aug  7 16:29 Techie_horizon/
-rw-rw-r--  1 vamshikumar vamshikumar   0 Aug  7 16:48 test1.txt
-rw-rw-r--  1 vamshikumar vamshikumar  12 Aug  7 16:46 test.txt
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ cp test.txt /home/vamshikumar/Desktop/Techie_horizon/
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ cd Techie_horizon/
vamshikumar@vamshikumar-virtual-machine:~/Desktop/Techie_horizon$ ll
total 12
drwxrwxr-x 2 vamshikumar vamshikumar 4096 Aug  7 16:52 ./
drwxr-xr-x 5 vamshikumar vamshikumar 4096 Aug  7 16:48 ../
-rw-rw-r-- 1 vamshikumar vamshikumar  12 Aug  7 16:52 test.txt
vamshikumar@vamshikumar-virtual-machine:~/Desktop/Techie_horizon$
```

9. Move a file to another location.

```
vamshikumar@vamshikumar-virtual-machine:~/Desktop$
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ ls
Techie_horizon  text.txt
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ cd Techie_horizon/
vamshikumar@vamshikumar-virtual-machine:~/Desktop/Techie_horizon$ ls
vamshikumar@vamshikumar-virtual-machine:~/Desktop/Techie_horizon$ cd ..
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ mv text.txt /home/vamshikumar/Desktop/Techie_horizon/
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ ls
Techie_horizon
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ cd Techie_horizon/
vamshikumar@vamshikumar-virtual-machine:~/Desktop/Techie_horizon$ ls
text.txt
vamshikumar@vamshikumar-virtual-machine:~/Desktop/Techie_horizon$
```

10. Rename a file.

```
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ ls
Techie_horizon
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ mv Techie_horizon Techie
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ ls
Techie
vamshikumar@vamshikumar-virtual-machine:~/Desktop$
```

11. Delete a file.

```
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ ls
Techie
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ rm -rf Techie
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ ls
vamshikumar@vamshikumar-virtual-machine:~/Desktop$
```

12. Grant or revoke permissions on a file or directory.

```
vamshikumar@vamshikumar-virtual-machine:~/Desktop/Techie_horizon$ ll
total 8
drwxrwxr-x 2 vamshikumar vamshikumar 4096 Aug  7 17:17 ./
drwxr-xr-x 3 vamshikumar vamshikumar 4096 Aug  7 17:17 ../
-rw-rw-r-- 1 vamshikumar vamshikumar   0 Aug  7 17:16 text.txt
vamshikumar@vamshikumar-virtual-machine:~/Desktop/Techie_horizon$ chmod 777 text.txt
chmod: cannot access 'text.txt': No such file or directory
vamshikumar@vamshikumar-virtual-machine:~/Desktop/Techie_horizon$ chmod 777 text.txt
vamshikumar@vamshikumar-virtual-machine:~/Desktop/Techie_horizon$ ll
total 8
drwxrwxr-x 2 vamshikumar vamshikumar 4096 Aug  7 17:17 ./
drwxr-xr-x 3 vamshikumar vamshikumar 4096 Aug  7 17:17 ../
-rwxrwxrwx 1 vamshikumar vamshikumar   0 Aug  7 17:16 text.txt*
vamshikumar@vamshikumar-virtual-machine:~/Desktop/Techie_horizon$ chmod 664 text.txt
vamshikumar@vamshikumar-virtual-machine:~/Desktop/Techie_horizon$ ll
total 8
drwxrwxr-x 2 vamshikumar vamshikumar 4096 Aug  7 17:17 ./
drwxr-xr-x 3 vamshikumar vamshikumar 4096 Aug  7 17:17 ../
-rw-rw-r-- 1 vamshikumar vamshikumar   0 Aug  7 17:16 text.txt
vamshikumar@vamshikumar-virtual-machine:~/Desktop/Techie_horizon$
```

13. View the current date and time.

```
vamshikumar@vamshikumar-virtual-machine:~/Desktop/Techie_horizon$ date
Friday 07 August 2026 05:18:58 PM IST
vamshikumar@vamshikumar-virtual-machine:~/Desktop/Techie_horizon$
```

14. Check the system uptime.

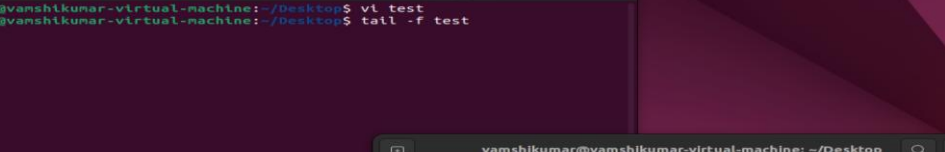
System uptime is the amount of time a computer or server has been **running continuously since its last boot (restart)**.

```
vamshikumar@vamshikumar-virtual-machine:~/Desktop/Techie_horizon$ uptime
17:19:41 up 29 min,  1 user,  load average: 0.00, 0.01, 0.04
vamshikumar@vamshikumar-virtual-machine:~/Desktop/Techie_horizon$
```

15. Search for files using the find command.

```
vamshikumar@vamshikumar-virtual-machine: ~/Desktop/Techie_horizon
vamshikumar@vamshikumar-virtual-machine:~/Desktop/Techie_horizon$ sudo find / -name sudoers
/etc/sudoers
/snap/core24/1643/etc/sudoers
/snap/core22/1612/etc/sudoers
find: '/run/user/1000/gvfs': Permission denied
find: '/run/user/1000/doc': Permission denied
/usr/share/doc/sudo/examples/sudoers
vamshikumar@vamshikumar-virtual-machine:~/Desktop/Techie_horizon$
```

16. Monitor log files using tail or grep.



The image displays two terminal windows from a Linux virtual machine. The top window shows the user 'vamshikumar' at the prompt 'vamshikumar@vamshikumar-virtual-machine: ~/Desktop'. They execute 'vi test' and then 'tail -f test'. The output shows 'hello' followed by a sequence of characters 'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j', 'k', 'l', 'm', 'n', 'o', 'p', 'q', 'r', 's', 't'. The bottom window shows the same user and prompt. They execute 'vi test' and then 'vi test' again. The second 'vi test' command results in a blank line.

```
vamshikumar@vamshikumar-virtual-machine: ~/Desktop
vamshikumar@vamshikumar-virtual-machine: ~/Desktop$ vi test
vamshikumar@vamshikumar-virtual-machine: ~/Desktop$ tail -f test
hello
a
b
c
d
e
f
g
h
i
j
k
l
m
n
o
p
q
r
s
t

vamshikumar@vamshikumar-virtual-machine: ~/Desktop
vamshikumar@vamshikumar-virtual-machine: ~/Desktop$ vi test
vamshikumar@vamshikumar-virtual-machine: ~/Desktop$ vi test
vamshikumar@vamshikumar-virtual-machine: ~/Desktop$
```

17. Print only specific lines from a file.



```
vamshikumar@vamshikumar-virtual-machine: ~/Desktop
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ cat test
hello
a
b
c
d
e
f
g
B
B
f
h
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ head -3 test
hello
a
b
vamshikumar@vamshikumar-virtual-machine:~/Desktop$ tail -4 test
B
B
f
h
vamshikumar@vamshikumar-virtual-machine:~/Desktop$
```